

Lab 3 – Linear Filters

The focus of this week's lab is on linear and non-linear filtering of digital images. In contrast to last week, rather than manipulating single pixels, we perform operations on neighborhoods of pixels.

Learning Objectives

- You can implement and apply simple linear filters in the spatial domain.
- You know how to deal with boundary pixels.
- You know which filters to use to sharpen an image.

1 Linear Shift-Invariant Filters

1.1 Lowpass Filter

In this exercise you will use two different lowpass filters and apply them to a number of digital images. Before applying the lowpass filter, briefly analyze (e.g, resolution, noise level, etc.) and visualize each image.

1. In a first step, create two 3×3 spatial low-pass filters: an averaging filter with impulse response h_M and a Gaussian filter ($\sigma = 0.85$) with impulse response h_G , where

$$h_M = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ and } h_G = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

2. Apply these filter kernels to the given images and compare the results between average and Gaussian filtering. Try changing the size of the filters.

Hints: A convenient way to apply the filter to the image is using the `convolve(...)`-command from the `scipy.ndimage` package.

1.2 Highpass Filter

An easy way to implement a high-pass filter is to subtract the lowpass filtered (blurred) version of an image from the original image.

1. Apply the lowpass filters to the given images.
2. Create a highpass filtered version of the original images by employing the previously mentioned subtraction procedure. Be careful with data types (under- and overflow). Hint: Try computing `numpy.uint8(5) - numpy.uint8(6)`.
3. What image features are enhanced by the highpass filter?

1.3 Image Sharpening

A traditional method for sharpening an image involves combining the original image with an edge-enhanced version of itself. Edge enhancement can, among others, be achieved with LAPLACE filters. There are various ways to implement a LAPLACE filter, and two possible implementations are given below:

$$h_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } h_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

1. Apply any of the two given filters to the original image.
2. Check how the filter changes the image.
3. Try to sharpen the original image by combining it with the filtered image.
4. Check if you obtain better results if you use Gaussian unsharp masking instead of the Laplace filter.

2 Median Filter

The image `cellsSandP.tif` contains salt-and-pepper noise.

1. Try to remove this noise with an averaging filter.
2. Now, apply a median filter instead. How does the computation time compare to that of a linear filter?